

TERRAFORM ASSIGNMENT-1

Terraform Workflow:

Write Configuration (Create Terraform configuration files (.tf) to define the desired infrastructure)



terraform init (Command for Initialize the working directory)



terraform validate (Validate configuration syntax and references)



terraform plan (Preview infrastructure changes)



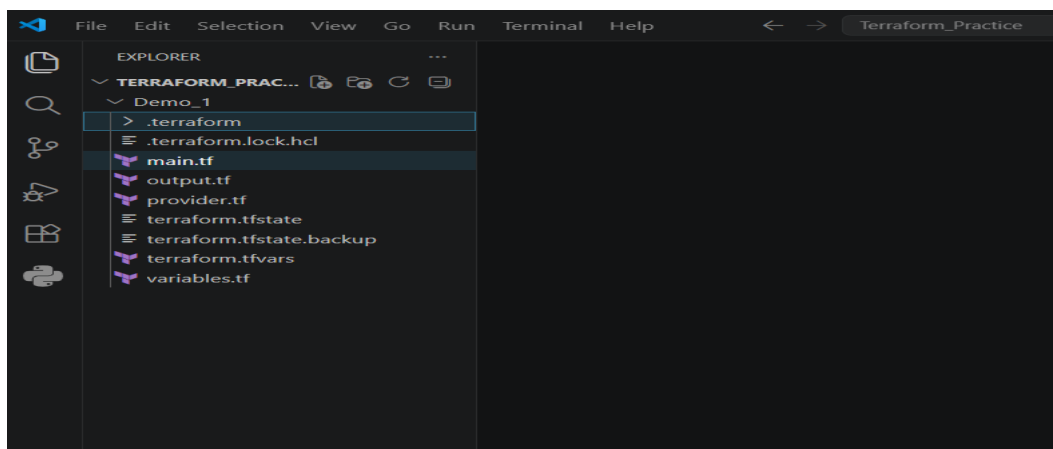
terraform apply (Creating or updating infrastructure)



terraform output (Display output values)



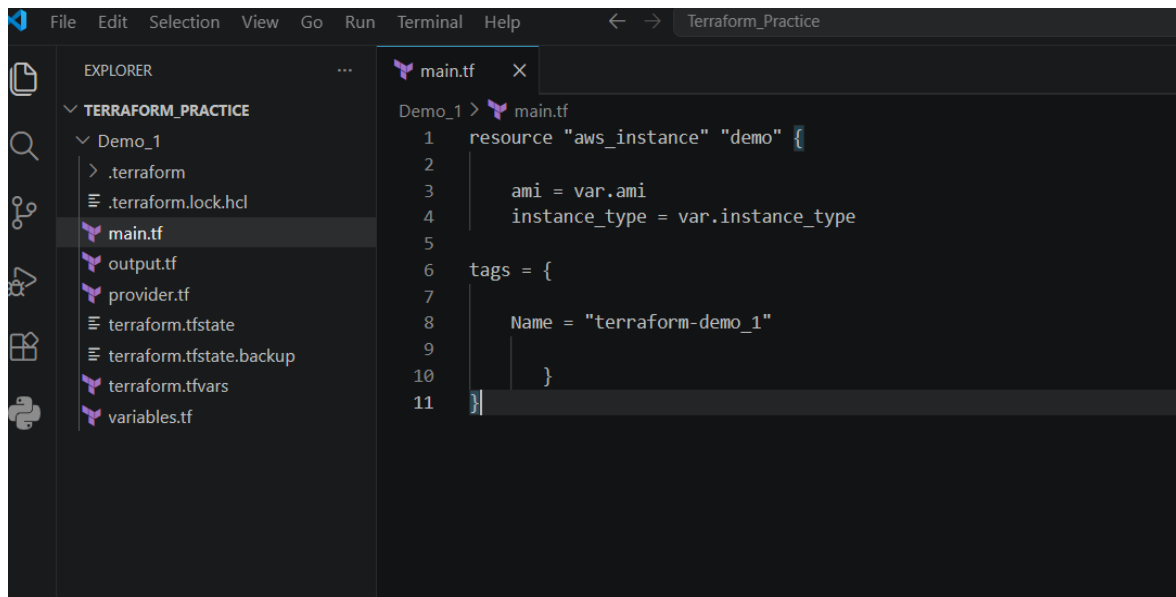
terraform destroy (Remove all managed infrastructure when no longer needed)



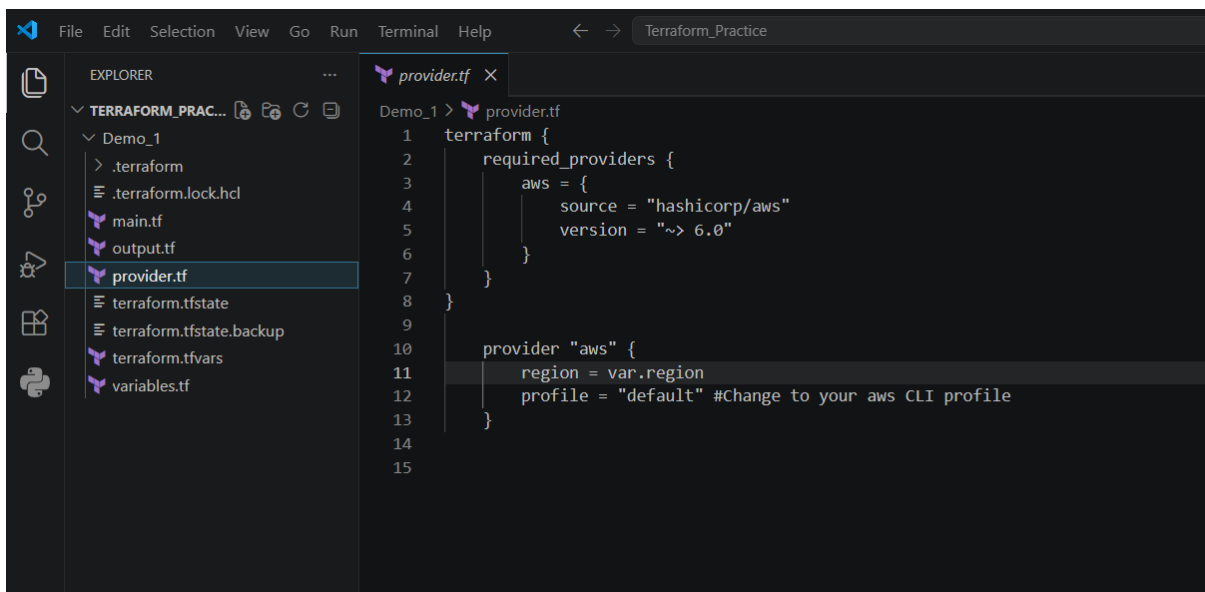
Common Terraform Files

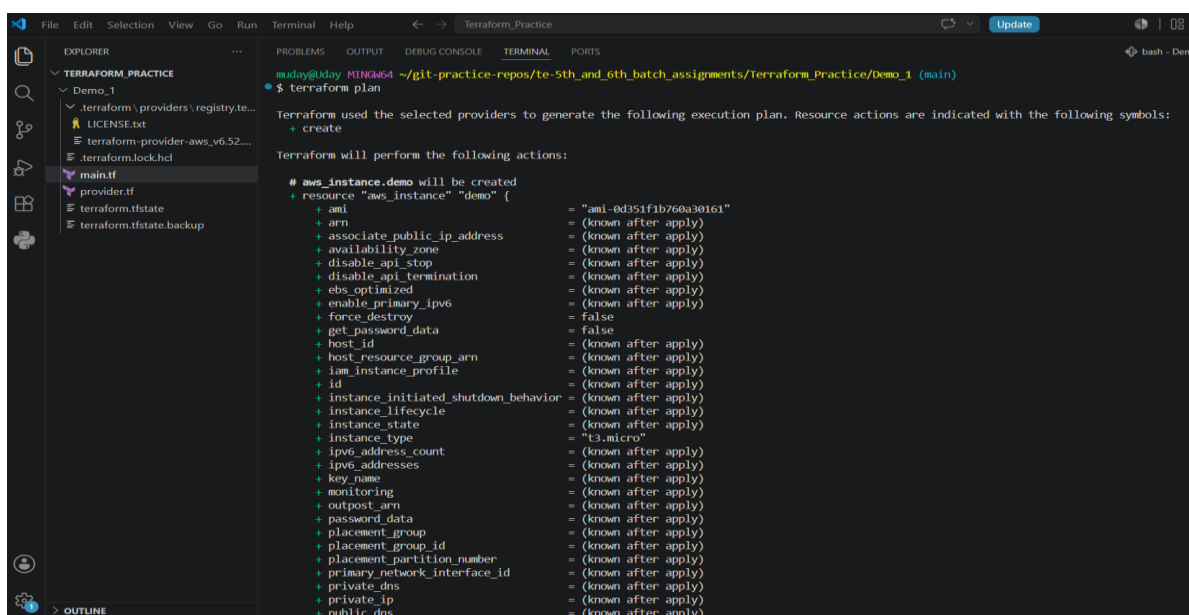
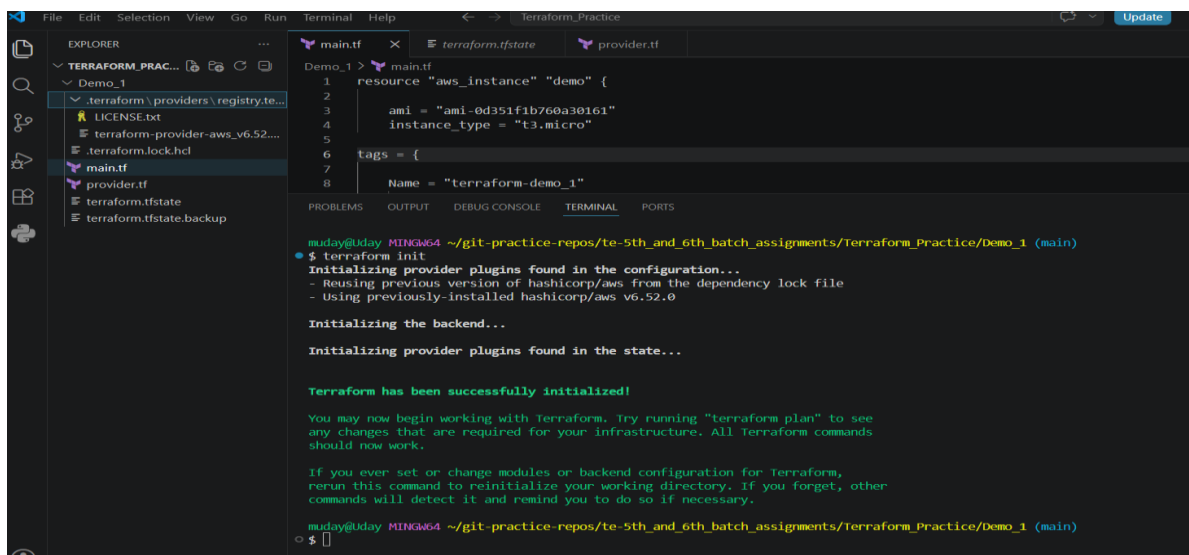
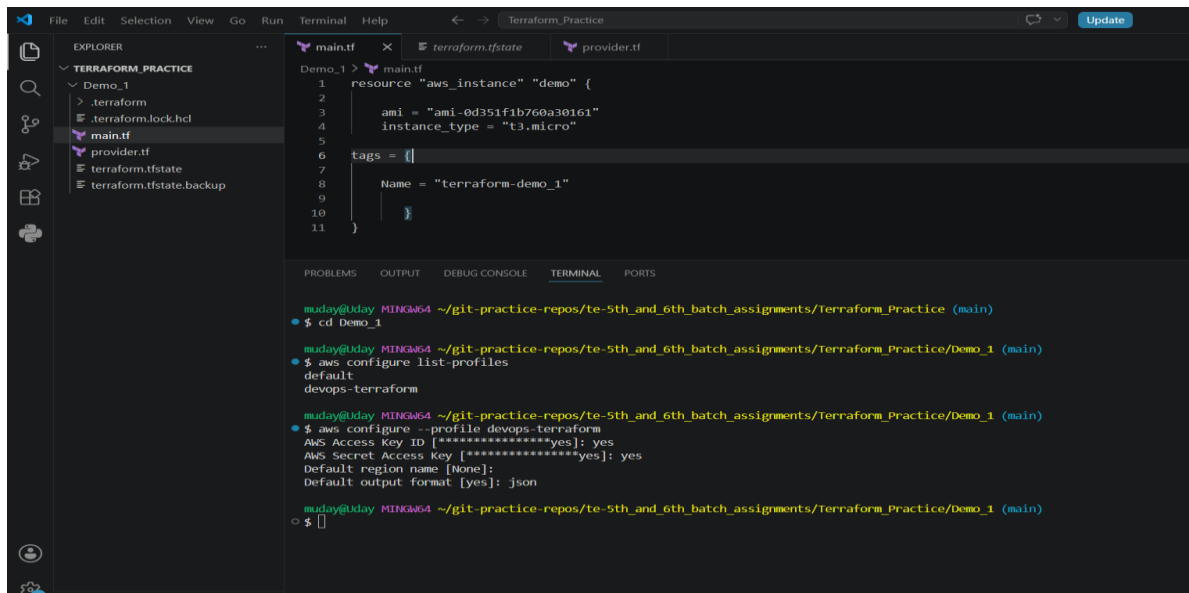
File	Purpose
main.tf	Main infrastructure resources
providers.tf	Provider configuration (AWS, Azure, GCP, etc.)
terraform.tfstate	Provides details of infrastructure
variables.tf	Defines input variables
terraform.tfvars	Assigns values to variables
outputs.tf	Defines output values
backend.tf	Configures remote state backend

main.tf file (Defines infrastructure resources (EC2, S3, VPC, etc.))



providers.tf (Configures the cloud provider (AWS, Azure, GCP))





```
main.tf terraform.tfstate provider.tf
Demo_1 > terraform.tfstate
1 {
2   "resources": [
3     {
4       "type": "aws_instance",
5       "name": "demo_1",
6       "provider": "aws",
7       "schema_version": 1,
8       "attributes": {
9         "availability_zone": "ap-south-1a",
10        "instance_type": "t3.micro",
11        "key_name": null,
12        "monitoring": false,
13        "subnet_id": "subnet-0a0c0621",
14        "tags": {
15          "Name": "demo_1"
16        },
17        "user_data": "#!/bin/bash\nsudo apt-get update\nsudo apt-get install -y\n\n",
18        "vpc_security_group_ids": ["sg-0a0c0621"]
19      },
20      "sensitive_attributes": [],
21      "private": {}
22    }
23  ],
24  "outputs": {}
25  }
26  }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform apply
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
+ secondary_network_interface (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.demo_1: Creating...
aws_instance.demo_1: Still creating... [00m10s elapsed]
aws_instance.demo_1: Creation complete after 13s [id=i-0202966b9575a6df0]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
terraform-demo_1	i-0e0ac06214f725630	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-232-1-126.ap-s...
terraform-demo_1	i-0202966b9575a6df0	Terminated	t3.micro	-	ap-south-1a	-

terraform output (Display output values)

```
main.tf terraform.tfstate output.tf provider.tf
Demo_1 > output.tf
1 #PUBLIC IP
2 output "public_ip" {
3   description = "Public IP of EC2 Instance"
4   value       = aws_instance.demo.public_ip
5 }
6 #AMI ID
7 output "ami" {
8   description = "AMI ID of EC2 Instance"
9   value       = aws_instance.demo.ami
10 }
11 #ARN
12 output "arn" {
13   description = "ARN of EC2 Instance"
14   value       = aws_instance.demo.arn
15 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform apply
Outputs:
ami = "ami-0d351f1b760a30161"
arn = "arn:aws:ec2:ap-south-1:504895205995:instance/i-0202966b9575a6df0"
public_ip = "3.108.223.73"

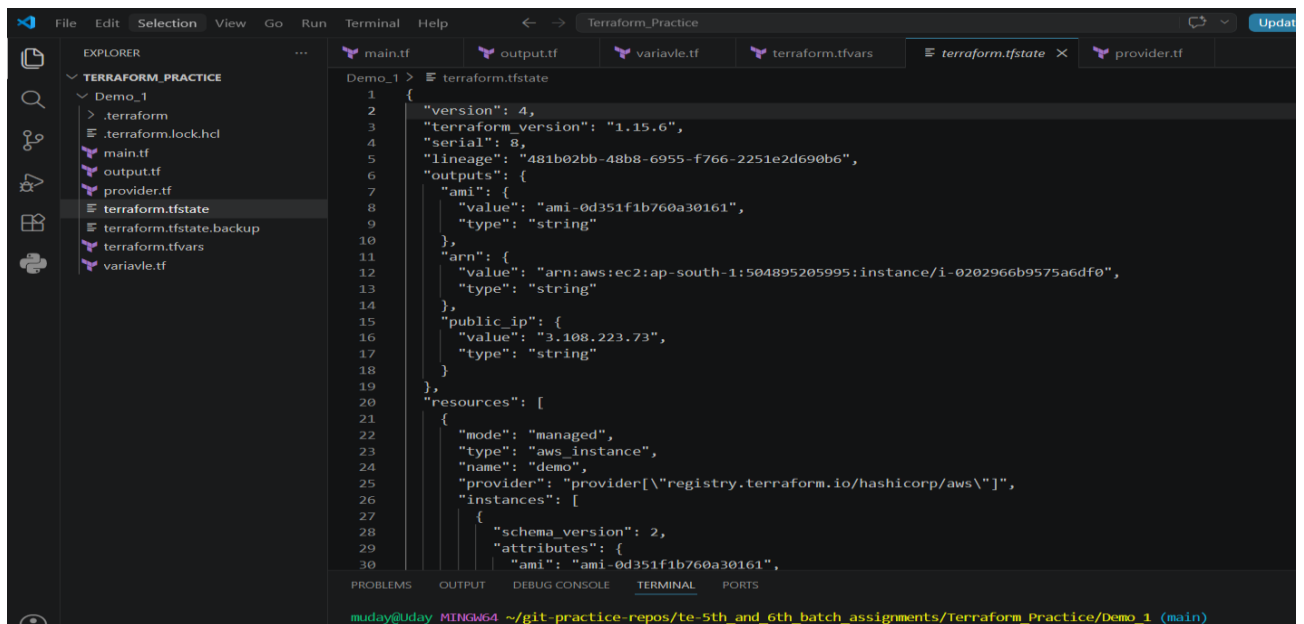
muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform output public_ip
"3.108.223.73"

muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform output ami
"ami-0d351f1b760a30161"

muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform output arn
"arn:aws:ec2:ap-south-1:504895205995:instance/i-0202966b9575a6df0"

muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
```

terraform State file (Provides details of infrastructure)

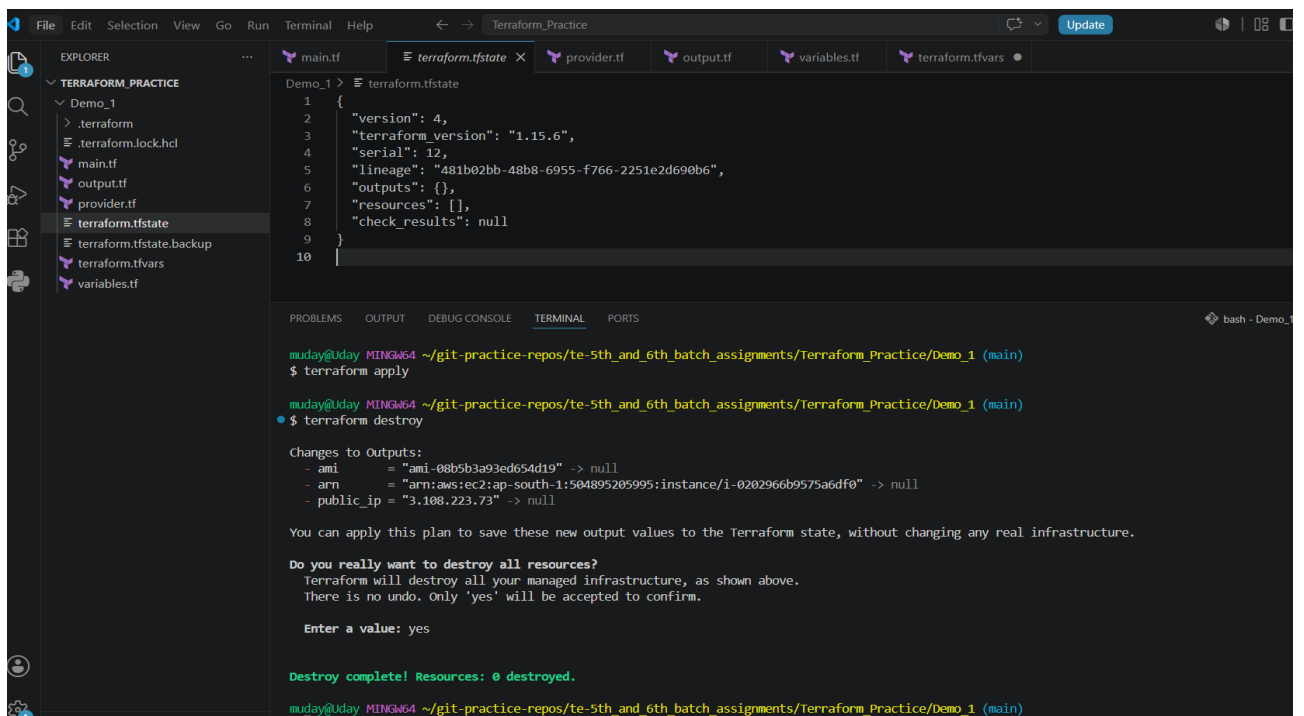


The screenshot shows the VS Code editor with the 'terraform.tfstate' file open. The file contains the following JSON structure:

```
1 {
2   "version": 4,
3   "terraform_version": "1.15.6",
4   "serial": 8,
5   "lineage": "481b02bb-48b8-6955-f766-2251e2d690b6",
6   "outputs": {
7     "ami": {
8       "value": "ami-0d351f1b760a30161",
9       "type": "string"
10    },
11    "arn": {
12      "value": "arn:aws:ec2:ap-south-1:504895205995:instance/i-0202966b9575a6df0",
13      "type": "string"
14    },
15    "public_ip": {
16      "value": "3.108.223.73",
17      "type": "string"
18    }
19  },
20  "resources": [
21    {
22      "mode": "managed",
23      "type": "aws_instance",
24      "name": "demo",
25      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
26      "instances": [
27        {
28          "schema_version": 2,
29          "attributes": {
30            "ami": "ami-0d351f1b760a30161",
```

The Explorer sidebar on the left shows the project structure for 'TERRAFORM_PRACTICE', including files like '.terraform', 'main.tf', 'output.tf', 'provider.tf', 'terraform.tfstate', 'terraform.tfstate.backup', 'terraform.tfvars', and 'variable.tf'.

Terraform Destroy(Remove all managed infrastructure)



The screenshot shows the VS Code editor with the 'terraform.tfstate' file open. The file content is as follows:

```
1 {
2   "version": 4,
3   "terraform_version": "1.15.6",
4   "serial": 12,
5   "lineage": "481b02bb-48b8-6955-f766-2251e2d690b6",
6   "outputs": {},
7   "resources": [],
8   "check_results": null
9 }
10
```

The Terminal window shows the following commands and output:

```
muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform apply

muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
$ terraform destroy

Changes to Outputs:
- ami = "ami-08b5b3a93ed654d19" -> null
- arn = "arn:aws:ec2:ap-south-1:504895205995:instance/i-0202966b9575a6df0" -> null
- public_ip = "3.108.223.73" -> null

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

Destroy complete! Resources: 0 destroyed.

muday@kday MINGW64 ~/git-practice-repos/te-5th_and_6th_batch_assignments/Terraform_Practice/Demo_1 (main)
```